# BLOOD AND LYMPHATIC

| | |
|---|---|
| **TB lymph node** | 15 |
| **Sarcoidosis (lymph node)** | 17 |
| **Metastatic deposits in lymph node** | 71 |
| **Hodgkin lymphoma** | Demo |
| **Non Hodgkin lymphoma(Burkitt's lymphoma)** | Demo |

# LYMPH NODES

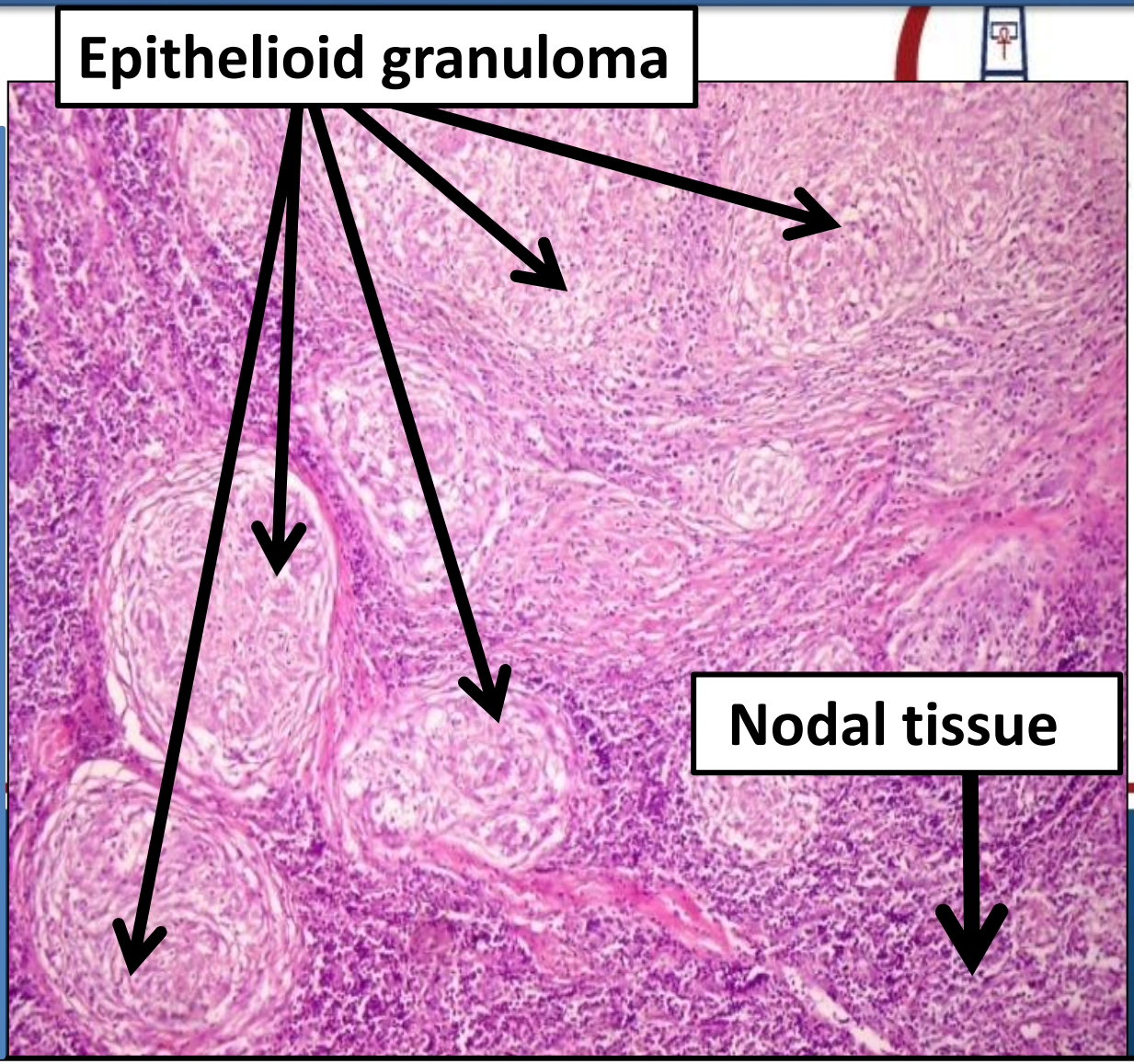
## Tuberculous Lymphadenitis (15)

**Clinical keynotes:**

1- In course of primary or secondary TB

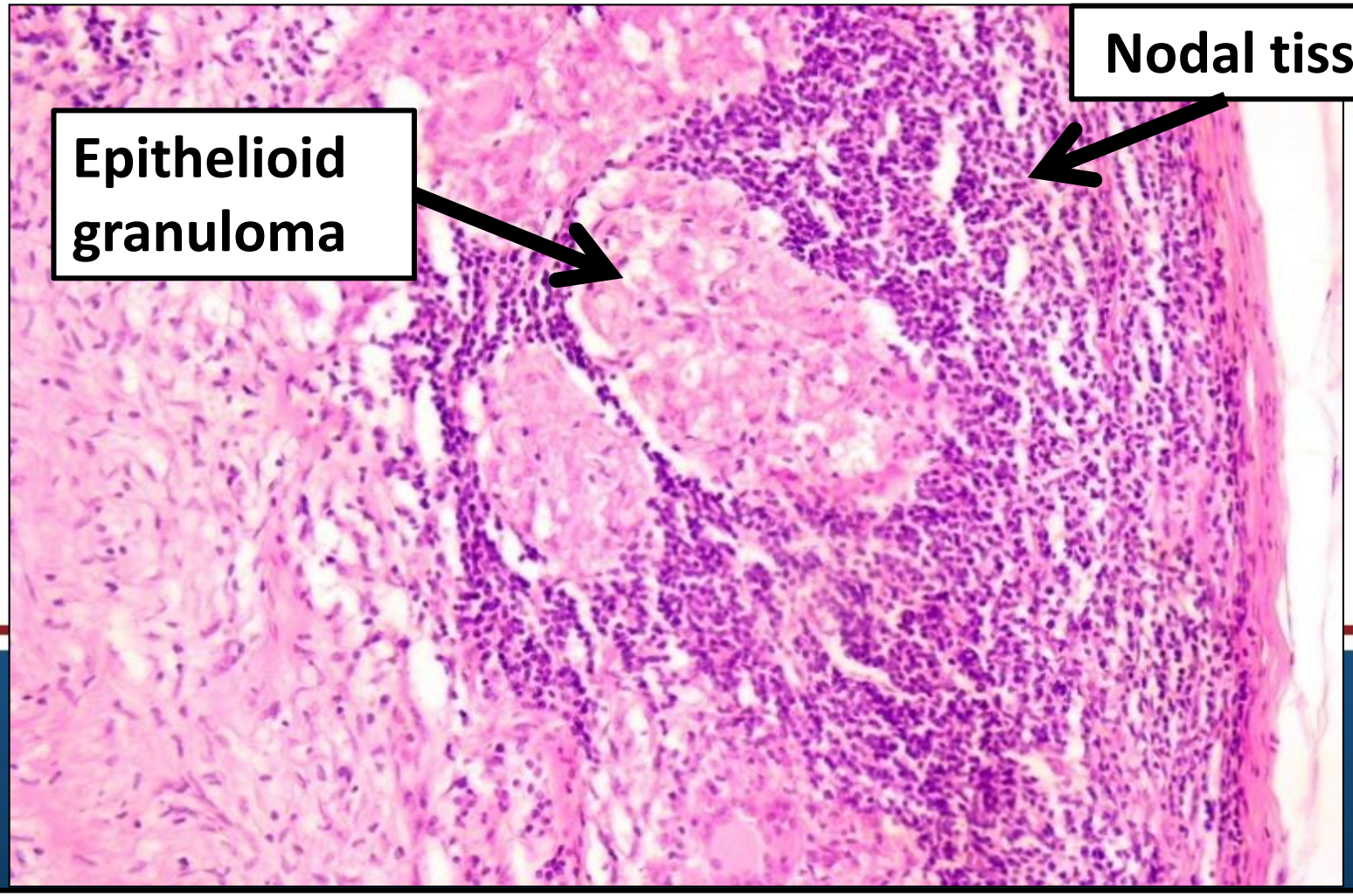
2- May affect cervical, hilar and mesenteric group of LNs especially in the primary type or complicating organ involvement in regional LNs of affected site

3- Confirmation by Zeihl-Neelsen Stain and bacteriological studies



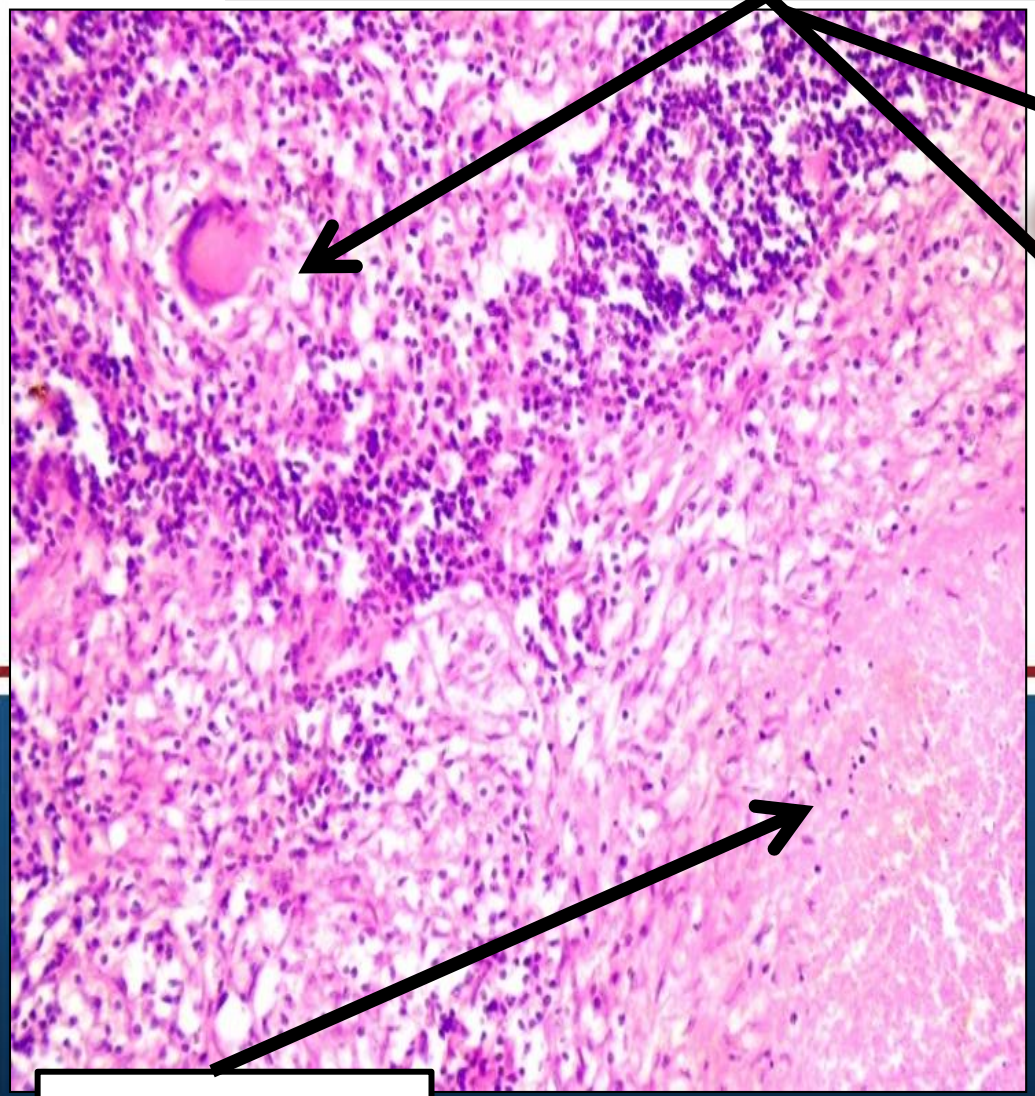

**Nodal architecture is replaced by multiple epithelioid large CONFLUENT CASEATING granulomas**

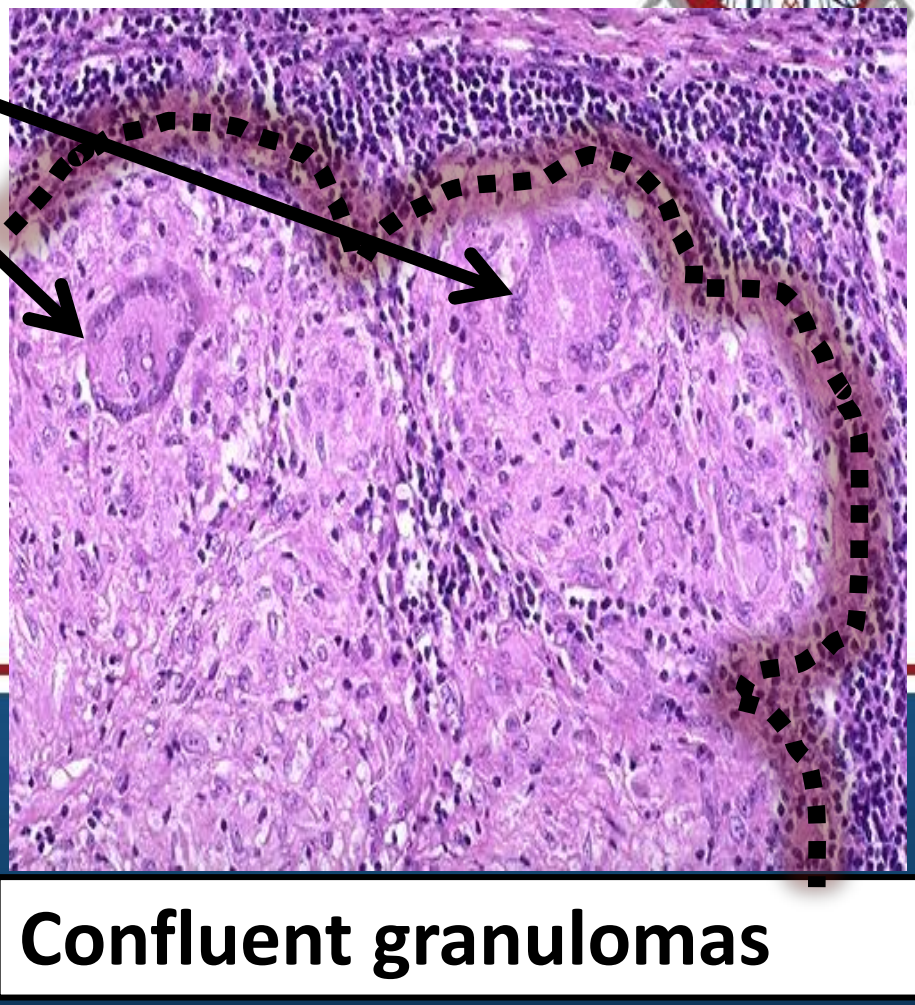
# LYMPH NODES Tuberculous Lymphadenitis (15)



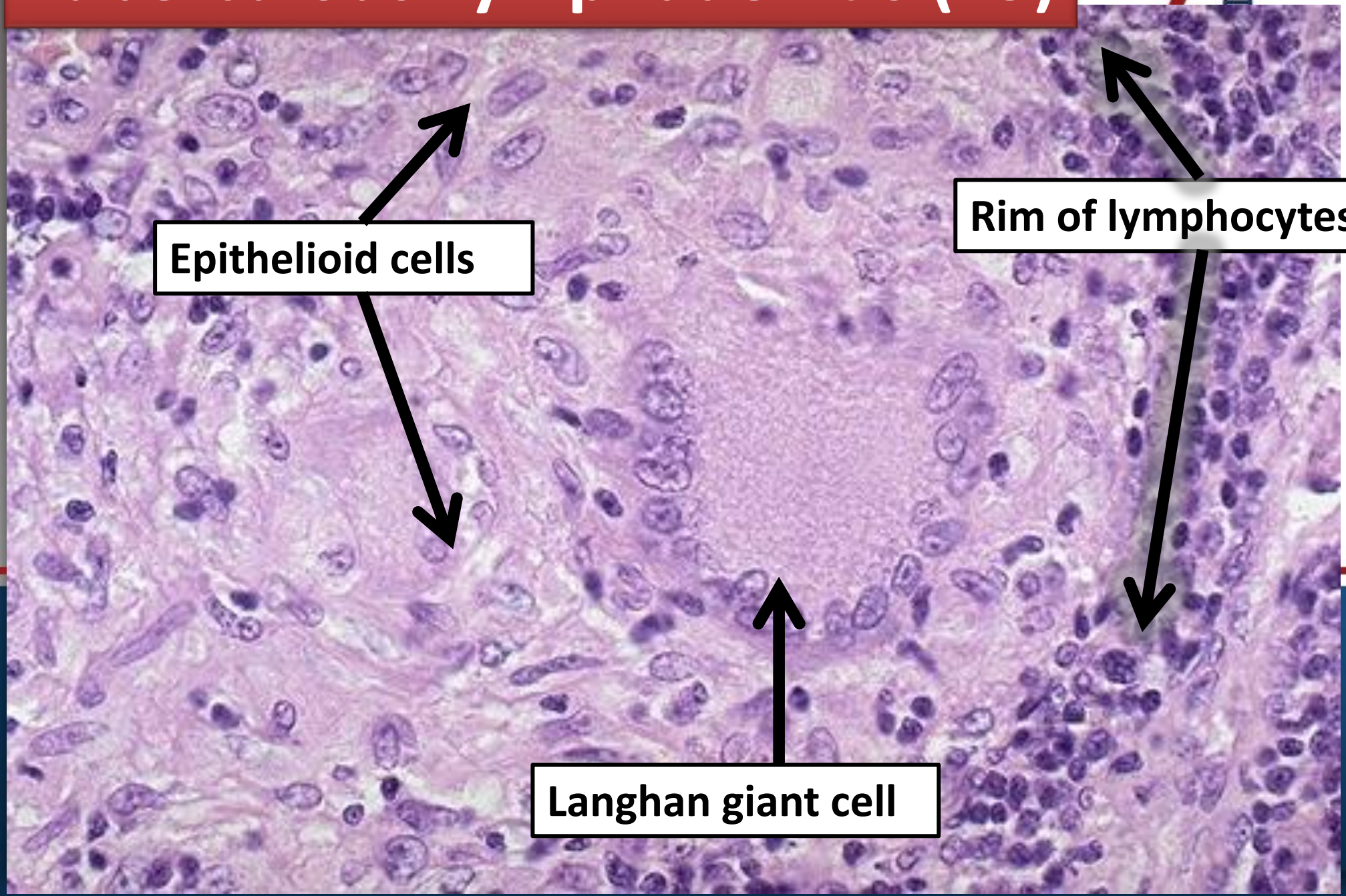

**Nature of this lesion: CHRONIC SPECIFIC INFLAMMATION of bacterial etiology-Necrotizing granulomatous lymphadenitis**

# Tuberculous Lymphadenitis (15)



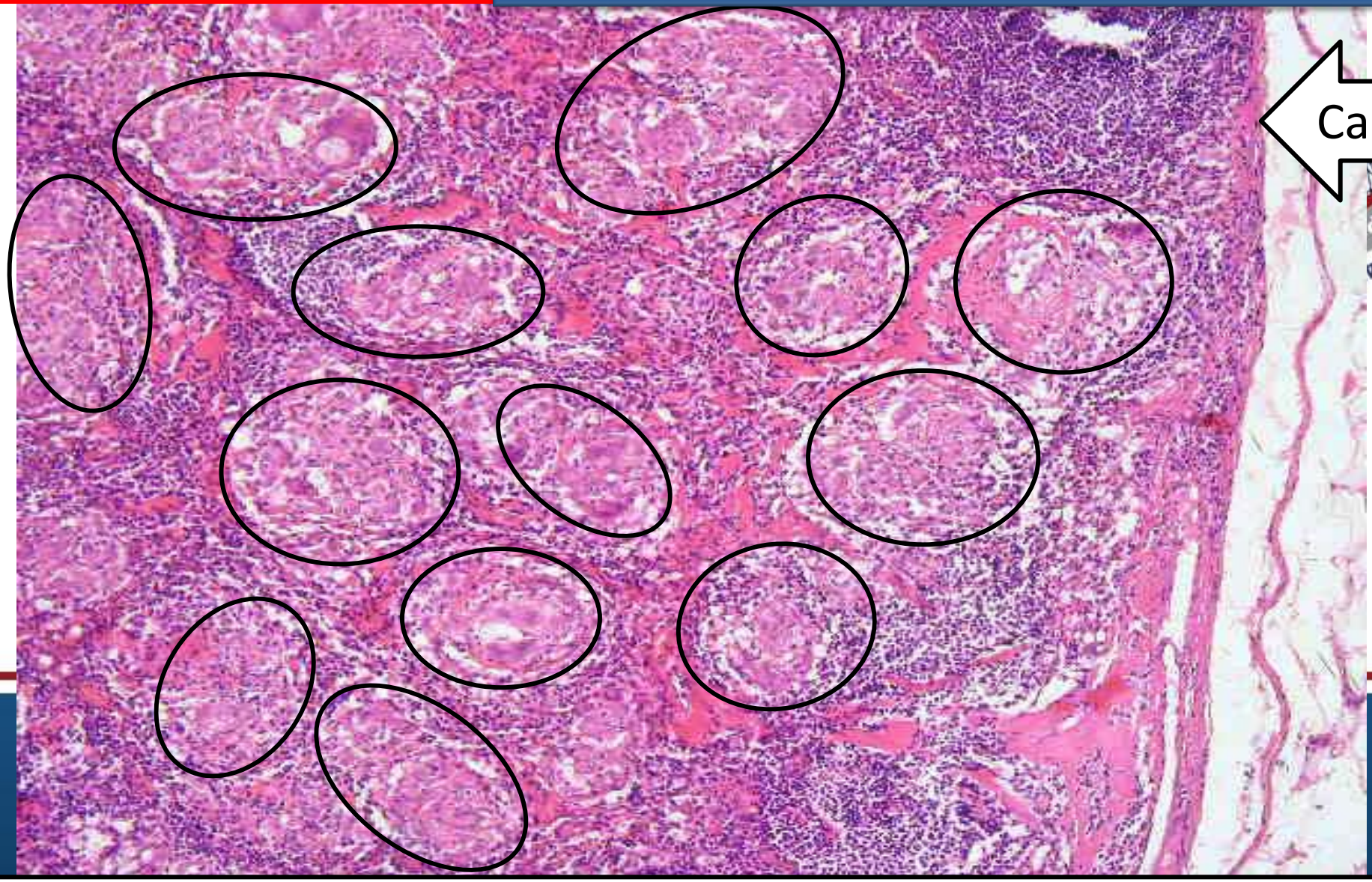



# Tuberculous Lymphadenitis (15)



# LYMPH NODES

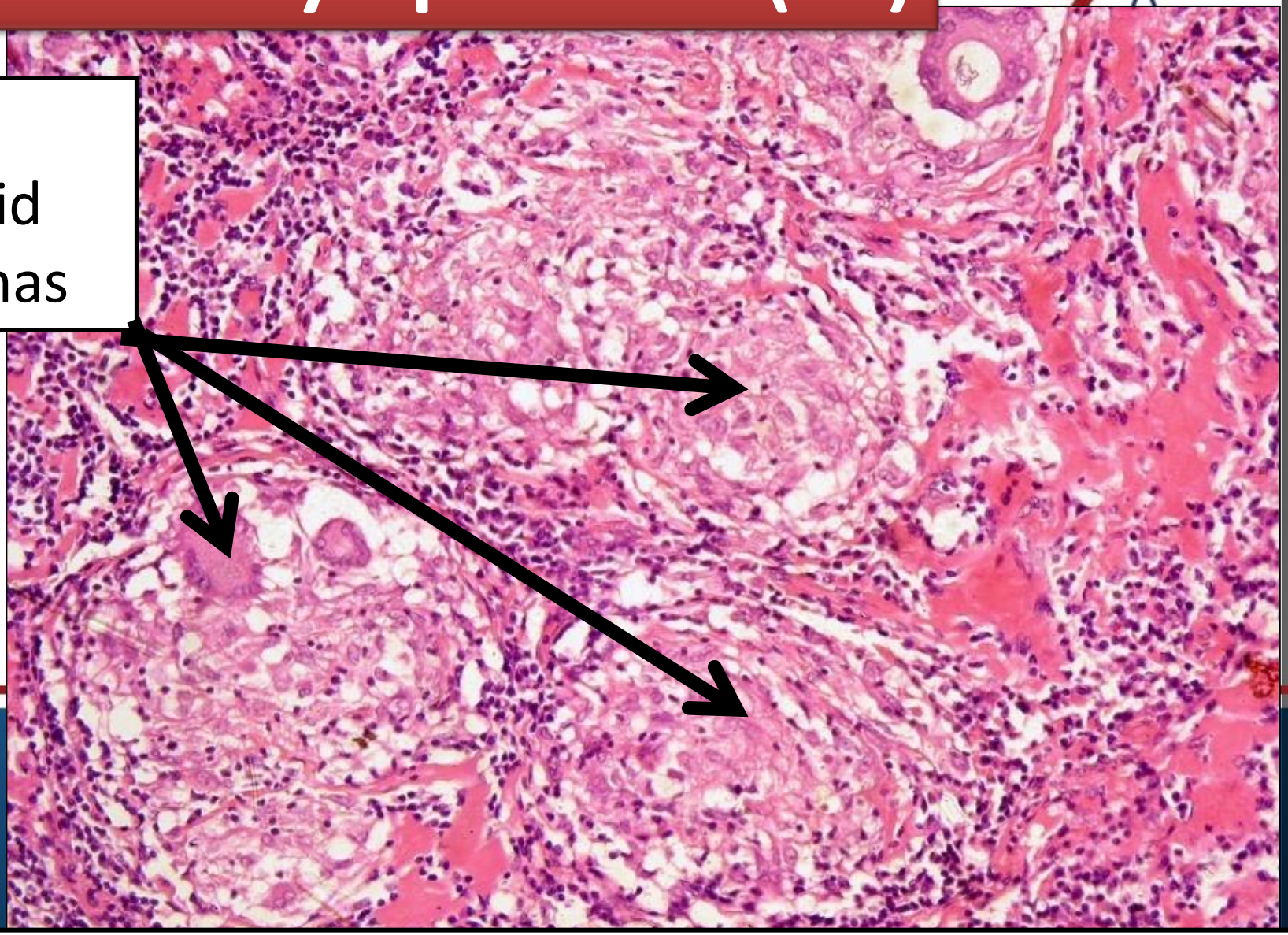
## Sarcoidosis of lymph nodes(17)




Replacement of nodal architecture by multiple **DISCRETE epithelioid NON caseating** granulomas

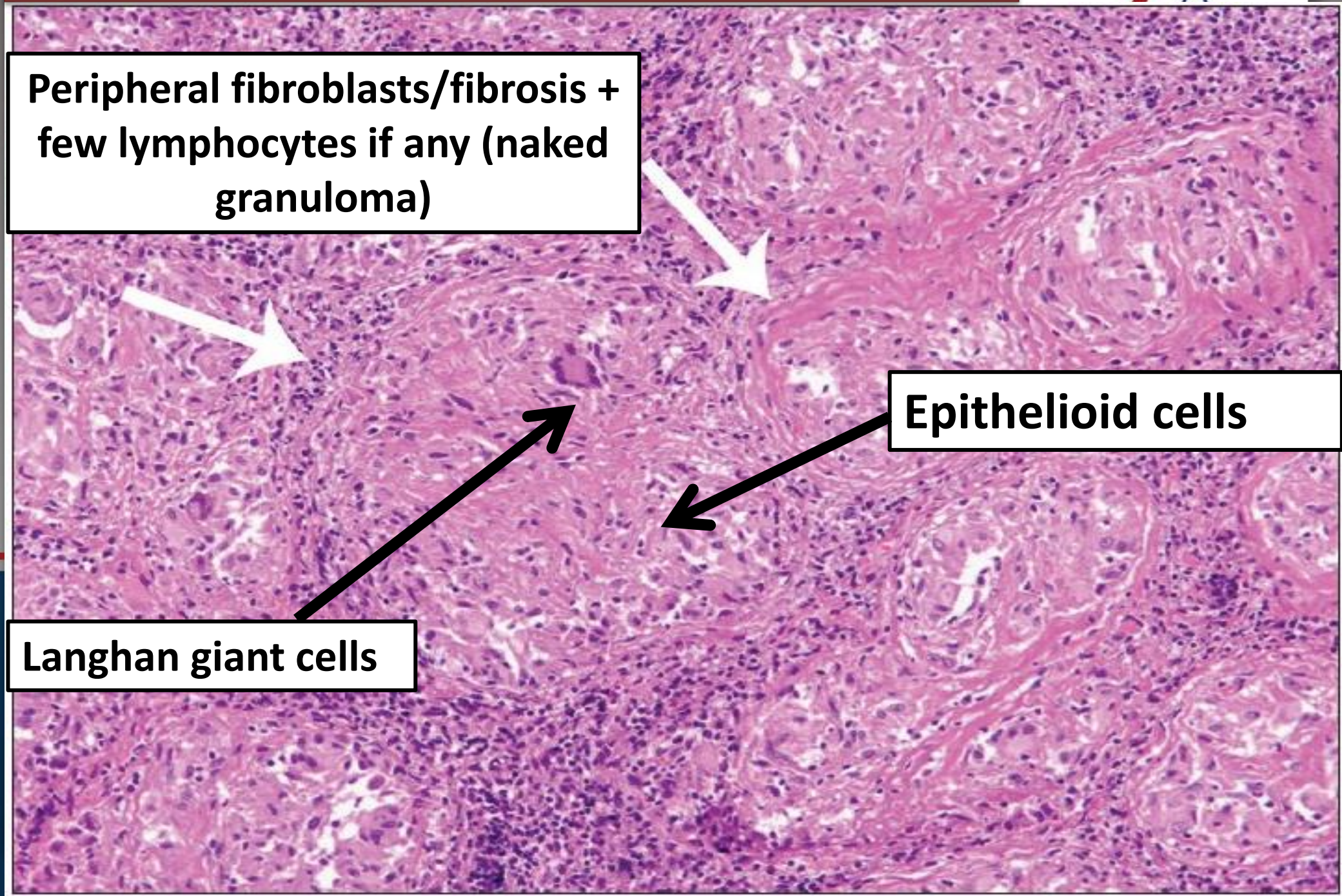
# Sarcoidosis of lymph nodes(17)

Multiple epithelioid granulomas



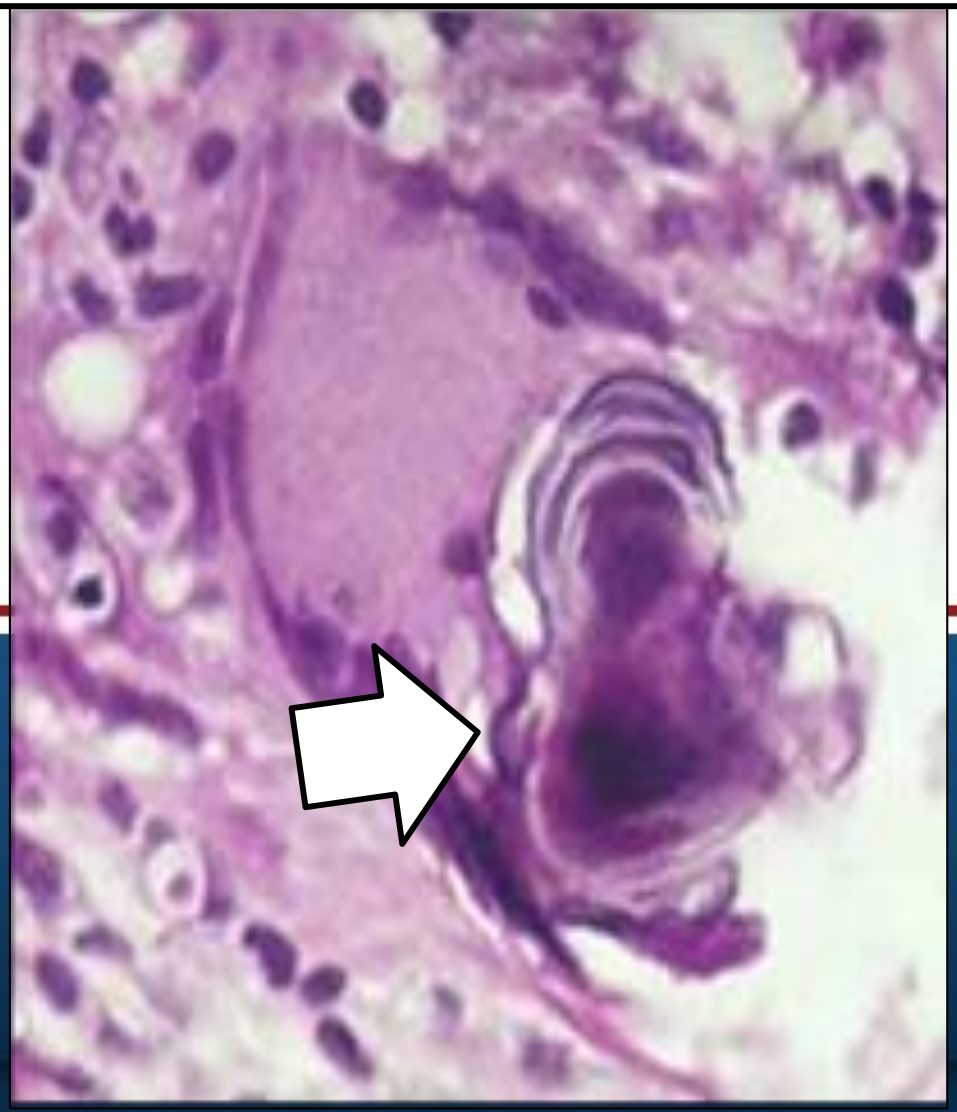
**Nature of this lesion: CHRONIC SPECIFIC INFLAMMATION of unknown etiology-Non necrotizing granulomatous lymphadenitis**

# Sarcoidosis of lymph nodes(17)



# Sarcoidosis of lymph nodes(17)

**Giant cell with Schaumann body (laminated calcified bodies)**



**Giant cell with Asteroid body (intracellular inclusions)**



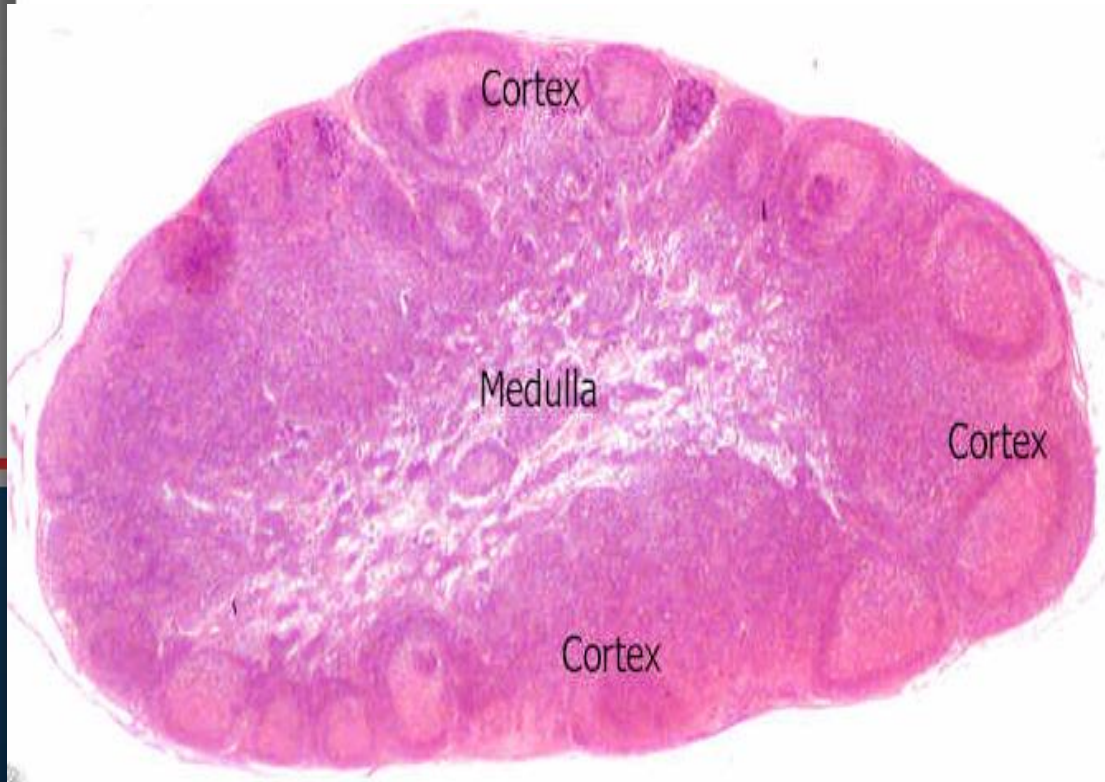
# LYMPH NODES

## Metastasis in lymph nodes (71)

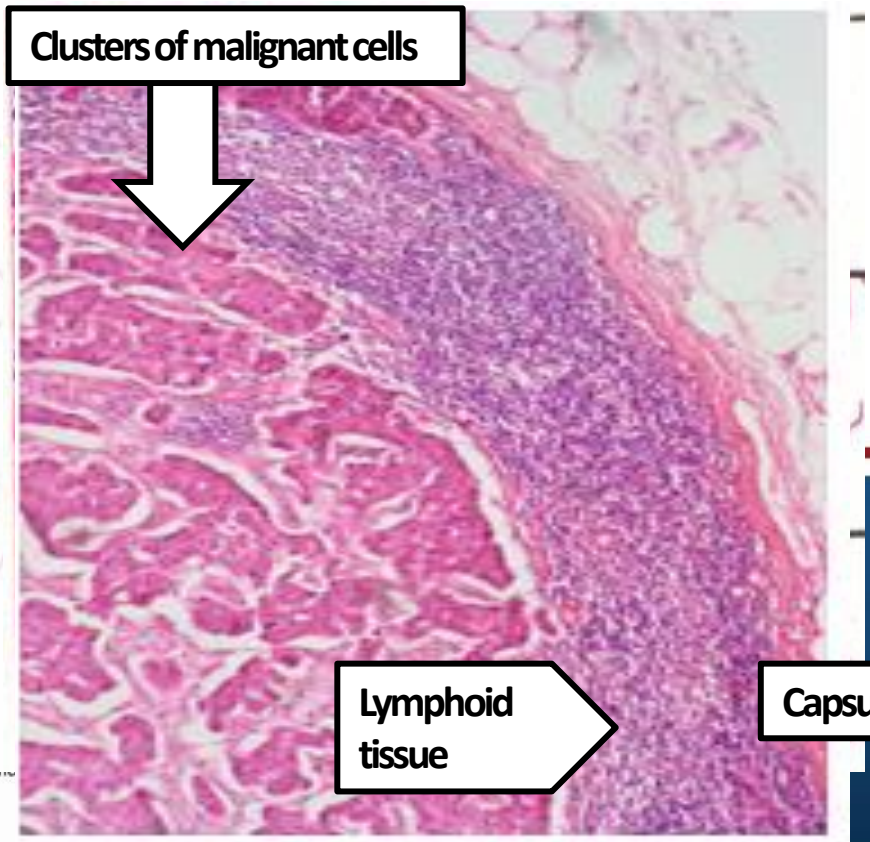
**Clinical keynotes:**

1- Represents a very important item in all tumor staging (T**N**M)

2- More common in carcinomas than in sarcomas

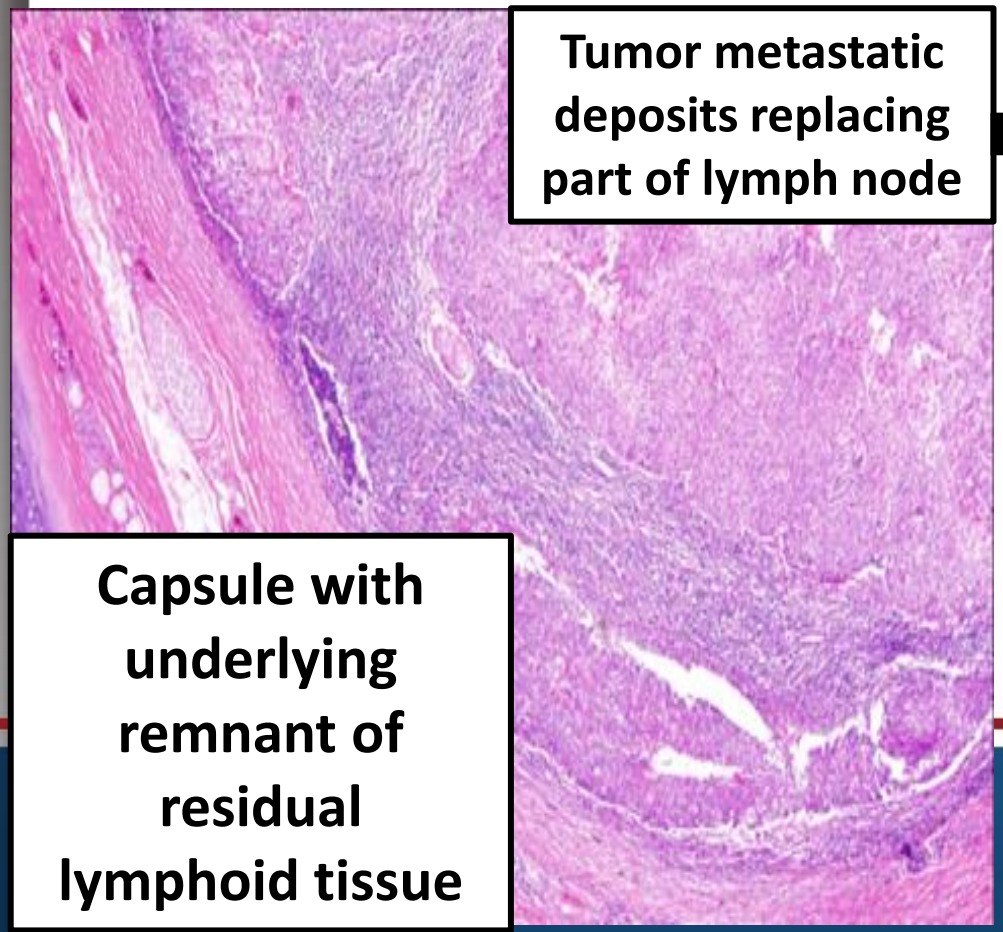



**Normal Lymph Node**



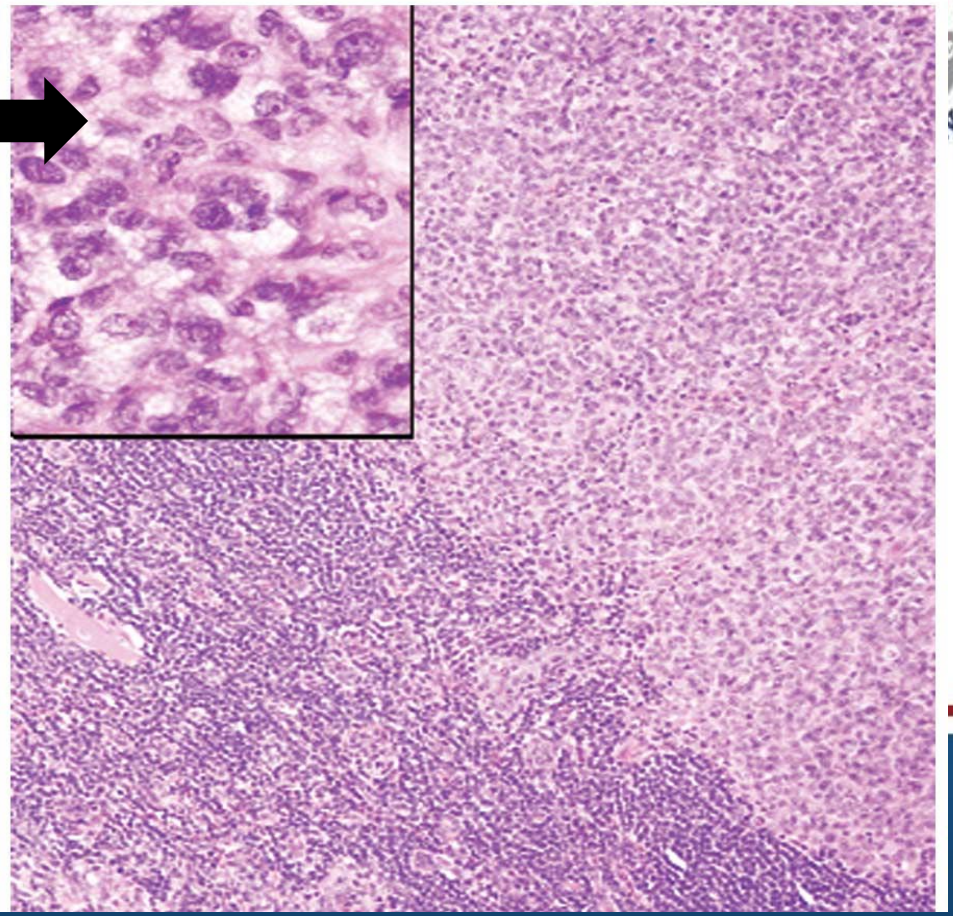

**Metastatic deposits**

# Metastasis in lymph nodes (71)



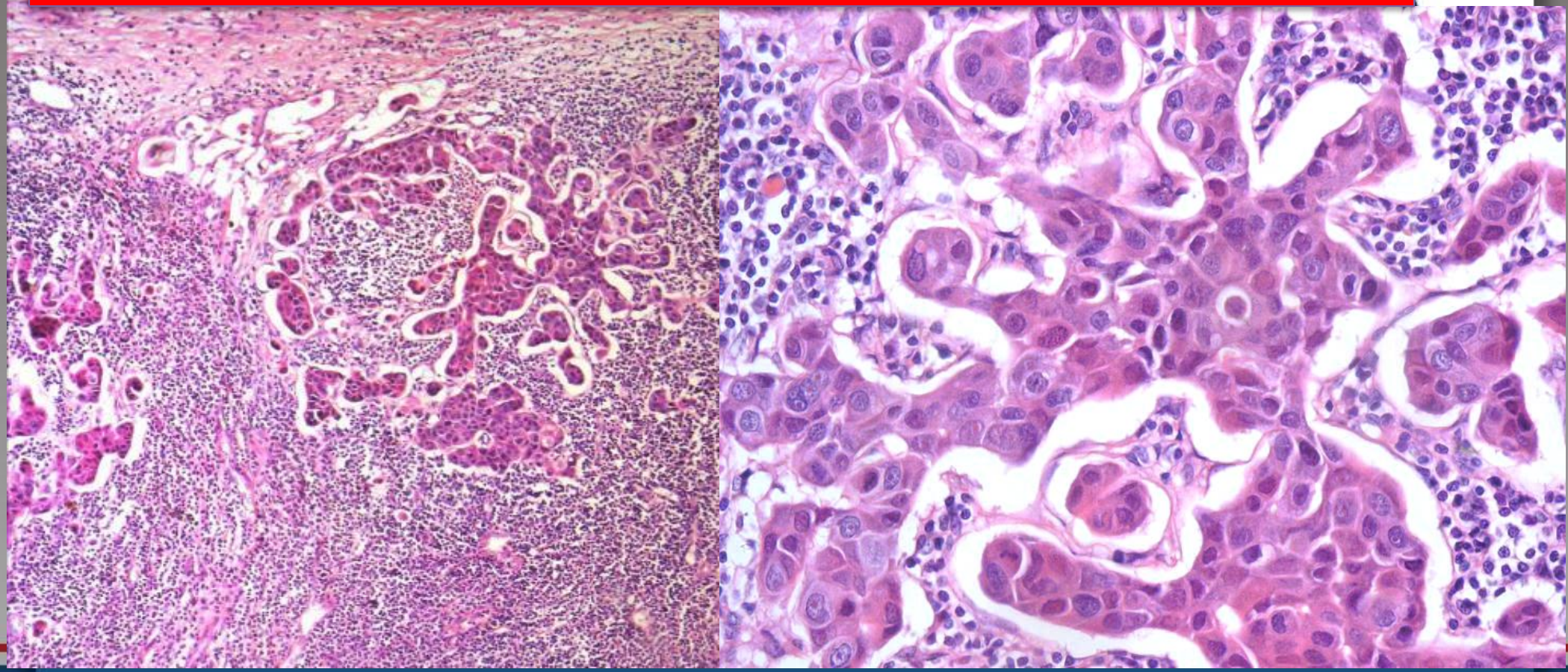
Capsule with underlying remnant of residual lymphoid tissue

Tumor metastatic deposits replacing part of lymph node



**Nature of this lesion: Spread of malignant tumors- mostly carcinomas**

# Metastatic deposits in lymph nodes(71)



**ARCHITECTURE OF METASTATIC tumor cells:**

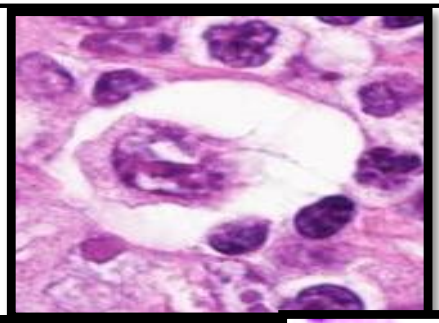
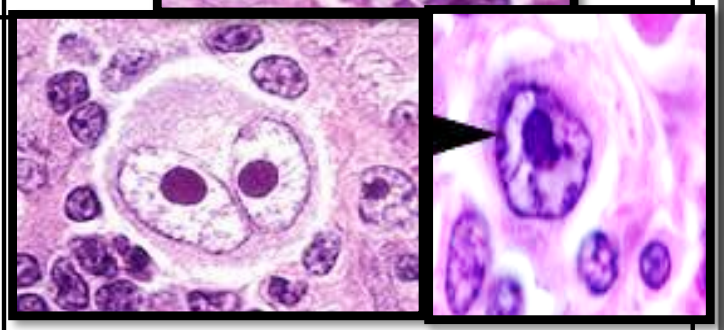
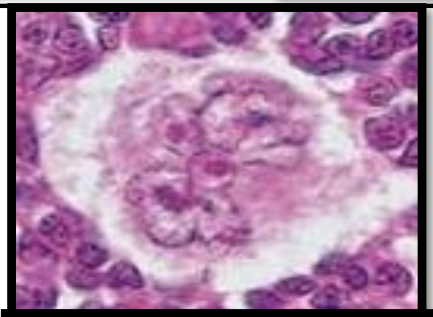
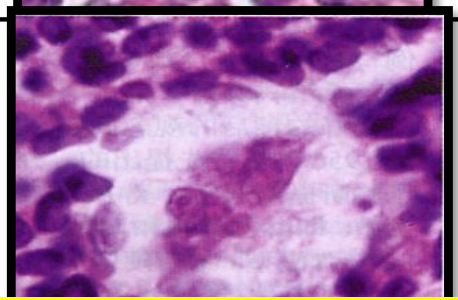
Papillary / Glandular/ Clusters / Sheets or nests ….(according to the tumor type)

**APPEARANCE OF METASTATIC tumor cells:**

Cells show pleomorphic hyperchromatic nuclei, mitotic activity

# LYMPH NODES

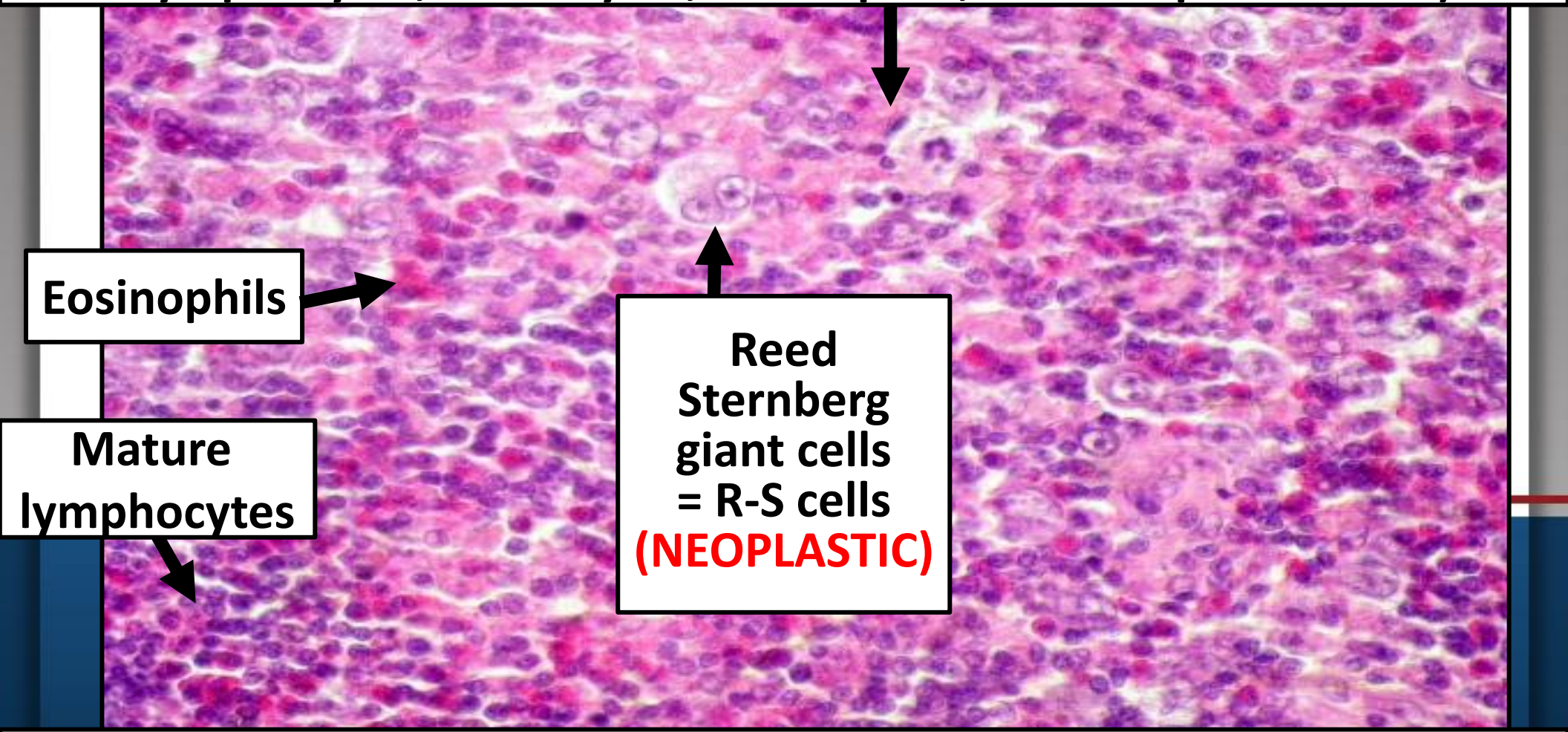
# Hodgkin Lymphoma (DEMO)

| Type | Histological Features | |
|---|---|---|
| ***Nodular sclerosis*** | Bands of fibrosis,<br>**Lacunar RSC** |  |
| ***Mixed cellularity*** | Composed of different cells.<br>**Classic RSC** |  |
| ***Lymphocyte predominance*** | Many lymphocytes<br>**Popcorn RSC** |  |
| ***Lymphocyte depletion*** | Many **Pleomorphic RSC** |  |

**NB : Differentiating individual type of R-S cells is not a subject of practical assessment**

# Mixed Cellularity Hodgkin Lymphoma (DEMO)

Polymorphic background (NON NEOPLASTIC=reactive population
: lymphocytes, histeocytes, eosinophils, PNL and plasma cells)




**Effacement or Loss of normal nodal architecture & replacement by a diffuse infiltrate**

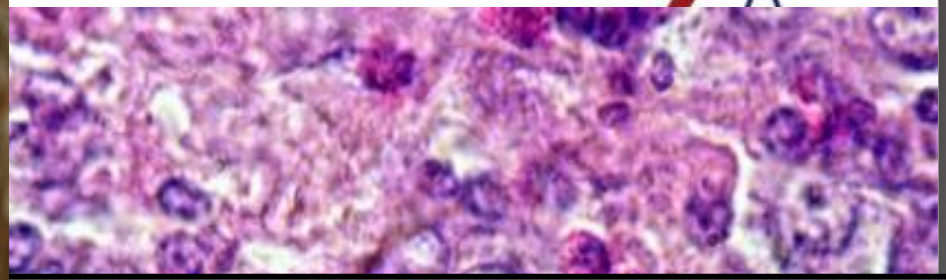
# Mixed Cellularity Hodgkin Lymphoma (DEMO)

Positive for CD15/

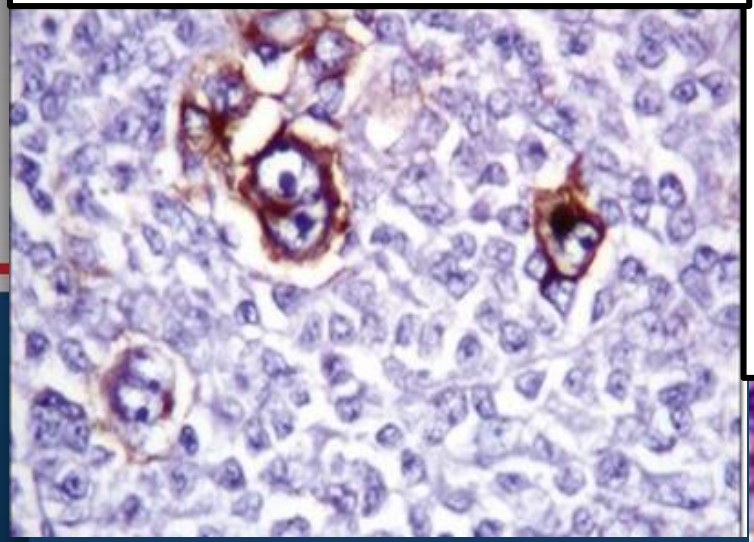
CD30 immuno-

histochemistry



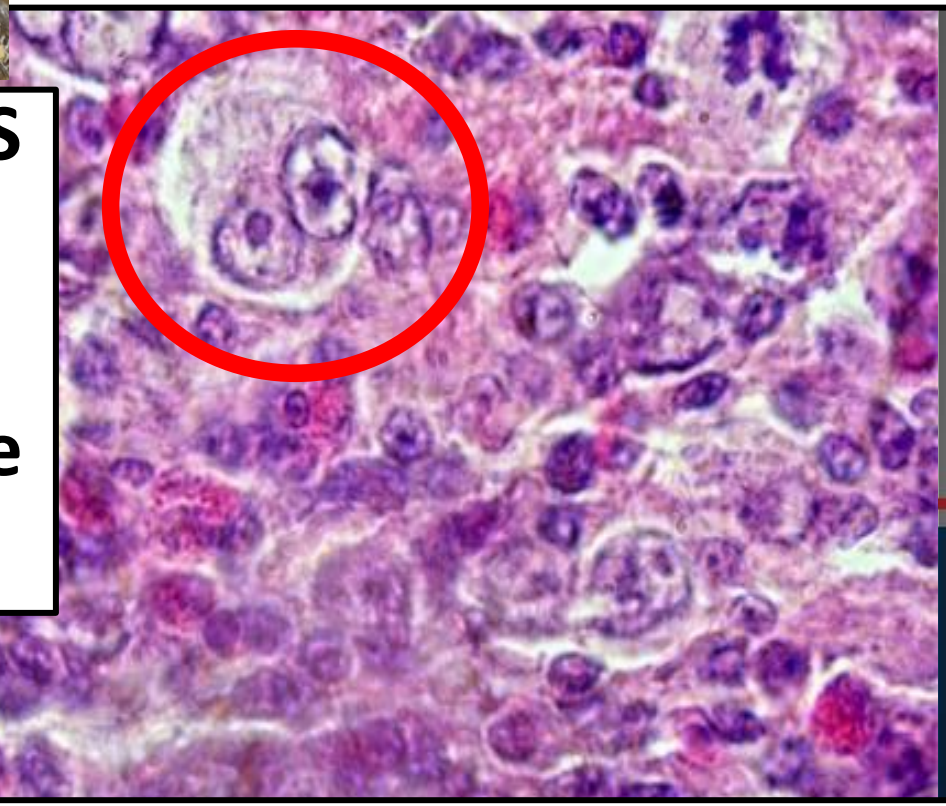


Prominent eosinophilic nucleoli



Classic R-S cells: MIRROR IMAGE

- Owl eye



Nature of this lesion: 1ry malignant tumors of lymph nodes: HODGKIN LYMPHOMA (NO benign counterpart in LNs)

# LYMPH NODES

# Non Hodgkin Lymphoma (DEMO)

WHO classification of selected types of non-Hodgkin lymphoma.

| B-cell lymphoma | T-cell lymphoma |
| --- | --- |
| **B CELL PRECURSOR** | **T CELL PRECURSOR** |
| • B-cell Leukemia/ lymphoma | • **T-cell Leukemia/lymphoma** |
| Mature B-cell Lymphomas | **Mature T-cell lymphomas** |
| • Follicular | • **Peripheral T cell lymphoma** |
| • Small lymphocytic | • **Anaplastic large cell (ALCL)** |
| • Lymphoplasmacytic | • **Mycosis fungoides** |
| • Marginal zone B-cell | |
| • Mantle cell | |
| • Diffuse large B-cell (DLBCL) | |
| • Burkitt's | |

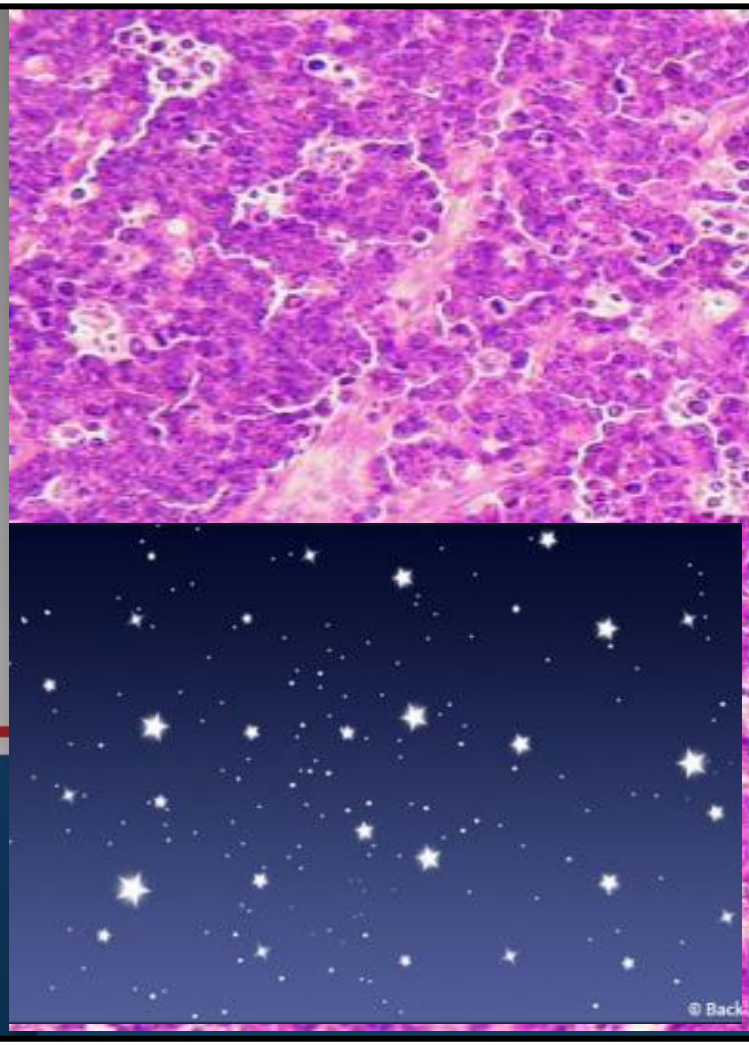
# Non Hodgkin Lymphoma: B cell Lymphoma Example: Burkitt lymphoma (DEMO)



- NON HODGKIN B-CELL LYMPHOMA
- AFRICAN CHILDREN
- JAW AFFECTION

# Burkitt lymphoma (DEMO)

**Monomorphic background infiltrate of immature lymphocytes(NEOPLASTIC) With** Starry-sky appearance

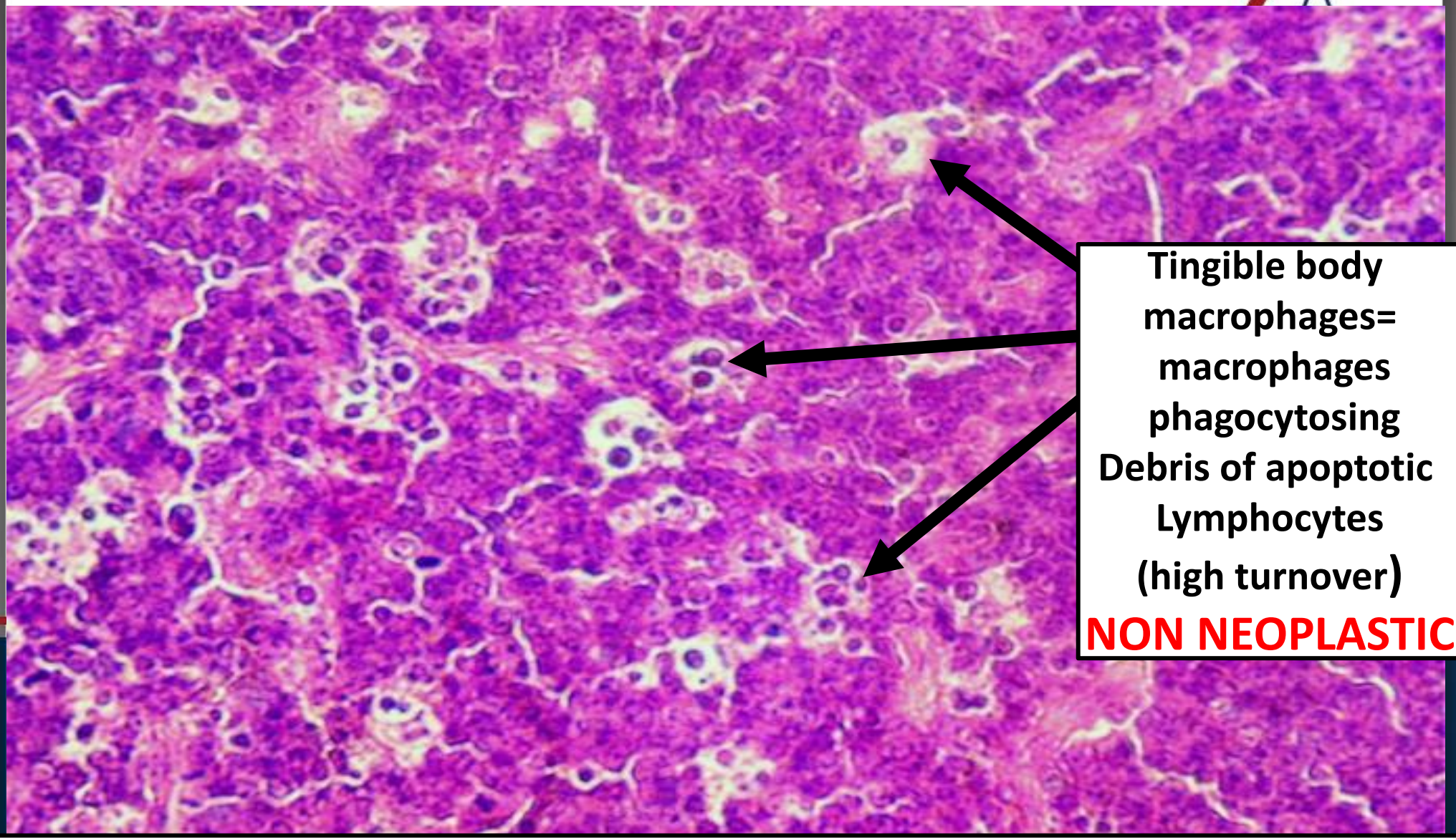


**Tingible body macrophages= macrophages phagocytosing Debri of apoptotic Lymphocytes (high turnover)**

**NON NEOPLASTIC**

**Effacement or Loss of normal nodal architecture & replacement by a diffuse infiltrate**

# Burkitt lymphoma (DEMO)




**Nature of this lesion: 1ry malignant tumors of lymph nodes**

**NON HODGKIN B –CELL LYMPHOMA**




# GOOD LUCK